Sequence 15

CONSOLIDATION AND TRANSITION TO YEAR 12

25 May–11 June 2027

I Review of calculation, functions and geometry

Example 15.1: Annual review – part A

1. Reduce $168/252$ to lowest terms; calculate $2^{-3} + 1/4$.
2. Factorise $x^2 - 6x + 9$, then solve $(x-3)^2 = 4$.
3. Solve $\dfrac{x-2}{x+1} \geq 0$.
4. For $f(x) = 4 - (x-1)^2$ on $[-2;3]$, give the maximum and the abscissae where $f(x) = 0$.
5. With $A(0;1)$, $B(4;3)$, $C(6;4)$, prove collinearity and give an equation of the line.

Solution:

1. $168/252 = 2/3$, and $2^{-3} + 1/4 = 1/8 + 2/8 = 3/8$. 2. $(x-3)^2$; $x-3 = \pm 2$, so $S = \{1;5\}$. 3. Forbidden -1, zero 2; signs $+, -, +$, so $S =]-\infty;-1[\cup [2;+\infty[$. 4. The square is positive or zero, so $f(x) \leq 4$, with equality at $x = 1$. The zeros are $x = -1, 3$, both in the domain. 5. $\overrightarrow{AB} = (4;2)$, $\overrightarrow{AC} = (6;3) = \frac{3}{2}\overrightarrow{AB}$. An equation is $x - 2y + 2 = 0$.

Teacher notes

Indicative marking: 10 points, 2 per question. For question 3, reserve one point for the domain and the endpoints. For question 5, ask for the conclusion after the calculation.

II Review of data, probability and modelling

Example 15.2: Annual review – part B

1. Calculate the mean and median of $2; 4; 4; 6; 9$. Determine Q_1, Q_3.
2. A price drops by $15\,\%$, then rises by $20\,\%$. Give the overall rate and the rate that then restores the initial price.
3. In a fictional population of 100 people, 40 play sport (S), 30 come by bike (V), 18 do both. Construct the two-way table and calculate $P_S(V)$, $P_V(S)$ for an equally likely draw.
4. For a rectangle of perimeter 24 m, express the area as a function of one dimension x and prove its maximum.

Solution:

1. Mean $25/5 = 5$, median 4. Quartile ranks 2 and 4, so $Q_1 = 4$, $Q_3 = 6$.
2. Overall multiplier $0.85 \times 1.20 = 1.02$: an increase of $2\,\%$. Reverse rate $1/1.02 - 1 = -1/51 \approx -1.96\,\%$.
3. Table: $S \cap V = 18$, $S \cap \overline{V} = 22$, $\overline{S} \cap V = 12$, $\overline{S} \cap \overline{V} = 48$. $P_S(V) = 18/40 = 45\,\%$, $P_V(S) = 18/30 = 60\,\%$.
4. $A(x) = x(12 - x) = 36 - (x - 6)^2$ on $]0; 12[$. The maximum is 36 m^2, attained for the square of side 6 m.

Teacher notes

Indicative marking: 10 points, respectively 2, 2, 3, 3. The review of parts A and B can be offered in one hour, then revisited with the course materials.